Geometry & Topology II: Course Notes

Mathilda Campillo

February 16, 2026

Remarks: These are the notes for Geometry and Topology II, Spring 2026. My personal comments will be in teal.

Contents

1 Vector bundles

1.1 Dual vector bundle

1.2 Cotangent bundle

Definition: The cotangent bundle is defined as

$$T^*M \equiv (TM)^*.$$

We call elements of this dual space, *covectors.* What we are doing here is just taking the dual vector space to T_pM at each point $p \in M$. As in the case of the tangent bundle, smooth local coordinates for M yield smooth local coordinates for its cotangent bundle

Recall: There is a one-to-one correspondence between between tangent vectors at p and derivations at p.

$$\{\text{vectors} \in T_pM\} \longleftrightarrow \{v : C^\infty(M) \to \mathbb{R} : v(fg)|_p = f|_p v(g) + g|_p v(f)\}$$

Notice that given a point $p \in M$, a covector $\omega \in T_p^*M$, and a tangent vector $v \in T_pM$, then one can define a pairing $\langle \omega, v\rangle \in \mathbb{R}$. This follows basically from the definition of dual vector space. Now, given a function f locally defined near $p \in M$ and a tangent vector $v \in T_pM$, one can similarly define a pairing given by $\langle df_p, v\rangle = v(f)_p \in T_p^*M$, where v is considered as a derivation defined near $p \in M$. We now give the precise definition of the differential.

Definition: Let $f : U \to \mathbb{R}$ be a smooth function defined on an open neighborhood of $p \in M$. Then the differential (of f at $p \in M$) is defined as

$$\begin{aligned} df_p : T_pM &\to T_p^*M \\ v &\mapsto df_p(v) \equiv v(f)_p \end{aligned}$$

The differential assigns a scalar to each tangent vector via the pairing in the previous comment.

1.2.1 How does df looks more concretely?

We can compute the differential in local coordinates as follows:

Given a local chart (U, x) near p, then each coordinate function x^i is smooth and has differential $dx^i|_p$. Moreover, the differentials $\{dx^1|_p, \dots, dx^n|_p\} \subset T_p^*M$ provide a dual basis to $\{\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p\} \subset T_pM$. In these local coordinates, we can write the differential as

$$df_p = \sum_i^n \frac{\partial f}{\partial x^i}\Big|_p dx^i\Big|_p$$

Notice that we can think of df as an analogue of the usual gradient, in a way that makes coordinate-independent sense on a manifold.

Lemma: $dx^i|_p, \dots, dx_n|_p$ is a basis for T_p^*M.

Some useful remarks:

1. Any covector $\omega_p \in T_p^*M$ can be written uniquely as

$$\omega_p = \omega_i dx^i\Big|_p, \quad \text{where } \omega_i = \omega\left(\frac{\partial}{\partial x^i}\Big|_p\right).$$

   This secretly says that all covectors come from differentials.

2. For each smooth function $f : M \to \mathbb{R}$, the map

$$\begin{aligned} df : M &\to T^*M \\ p &\mapsto df_p \in T_p^*M \end{aligned}$$

   is a smooth section. Why? because at each point we are choosing the linear functional $df_p : T_pM \to \mathbb{R}$. Now, we will later see that these sections will define something called covector fields or differential 1-forms.

Properties: Let $f, g \in C^\infty(M)$. Then

1. $d(af + bg) = adf + bdg$
2. $d(fg) = gdf + fdg$
3. $F : N \to M$ smooth map between smooth manifolds,

$$d(f \circ F)(V) = df(DF(v)) \forall v \in T_qN, q \in N.$$

Slightly off topic sidenote: We know that geometrically, we can think of a vector field on M as a rule that assigns an arrow to each point of M, but what kind of geometric picture can we form of a covector field?

The take away idea is that a nonzero linear functional $\omega_p \in T_p^*M$ is completely determined by two things:

1. Its kernel, which is a codimension one linear subspace of T_pM
2. The set of vectors in T_pM such that $\omega_p(v) = 1$, which determines an affine hyperplane parallel to its kernel.

Then, we can visualize a covector field as a defining pair of hyperplanes in each tangent space, one through the origin and another parallel to it, varying continuously from point to point.

Notice that up to this point, we have defined two different kinds of derivatives of f at a point $p \in M$:

1. The <u>pushforward</u> $f_* : T_pM \to T_{f(p)}\mathbb{R}$.
2. The <u>differential</u> $df_p : T_pM \to \mathbb{R}$.

But these are the same locally, once we take into account the canonical identification between $\mathbb{R}$ and its tangent space at any point.

Lemma: $f \in C^\infty(M)$, $f : M \to \mathbb{R}$ smooth, $\gamma : [0,1] \to M$ smooth path. Then

$$F(\gamma(1)) - F(\gamma(0)) = \int_0^1 df(\gamma'(t))dt.$$

Definition: A 1-form is a smooth section ω of T^*M. If $F : N \to M$ is a smooth function, then the pullback of ω is $F^*\omega$ on N, satisfying

$$F^*\omega(v) = \omega(DF(v)), \ \ \forall v \in T_pM.$$

Definition: $\gamma : [a,b] \to M$ be a smooth map. The line integral of ω along γ is

$$\int_a^b \omega(\gamma'(t))dt.$$

In general: $\omega|_p = \sum \omega|_p \frac{\partial}{\partial x_i} dx^i$

Examples: (1) $M = \mathbb{R}^2$ and consider $\omega = xdy$ which is not exact (not even close). Take the path $\gamma : [0,\pi] \to \mathbb{R}^2$ given by $\gamma(t) = (\cos(t), sin(t)$. Then

$$\gamma'(t) = \sin(t)\frac{\partial}{\partial x} + \cos(t)\frac{\partial}{\partial y}$$

$$\omega(\gamma'(t)) = \cos^2(t).$$

$$\int_0^1 \omega(\gamma'(t))dt = \int_0^1 \cos^2(t)dt > 0$$

Now take the path $\Gamma : [0,\pi] \to \mathbb{R}^2$ defined by $\Gamma(t) = (\frac{t}{\pi}, 0)$